

Assessment of Personal Direct-Reading Dust Monitors for the Measurement of Airborne Inhalable Dust

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The performances of five portable direct-reading dust monitors were investigated in a wind tunnel for a range of industrial dusts and three sizes of aluminium oxide test dust to mainly determine their suitability for measuring the inhalable fraction of airborne dust in workplaces. The instruments tested were Split 2 (SKC Ltd), Sidepak (TSI Inc.), Dataram (Thermo Electron Ltd), PDS-2 (Sibata Scientific Technology Ltd) and the Respicon TM (Hund Ltd). The instruments' responses were compared with reference dust samplers. These were the IOM sampler for the inhalable fraction and the Casella cyclone sampler for the respirable fraction. All instruments are predominantly responsive to and are designed to measure particles in the respirable size range, although two of the instruments, the Split 2 and Respicon TM, are claimed to be capable of measuring inhalable-sized particles. For the purpose of the tests, major modifications to an existing wind tunnel dust injection system were made to facilitate the generation of uniform concentrations of large inhalable-sized dust particles at low air velocities. Each monitor greatly underestimated the measurement of inhalable concentration for all the dusts tested, although the linearity was good over a wide range of concentrations for any particular size distribution of dust. However, their calibration factors, defined as the ratio of reference inhalable concentration to monitor concentration, were especially sensitive to changes in particle size as the response of the instruments decreased rapidly with increasing particle size. The monitors generally overestimated the measurement of respirable dust concentration by up to a factor of about 2, apart from the PDS-2, which underestimated it by a factor of up to 3. There was, however, a great deal more scatter in the reference respirable concentration measurements owing to the collection of small dust samples. Therefore, monitor linearity and effects of monitor response to changes in particle size could not be accurately investigated for the respirable fraction. The sampling head of the Split 2 monitor incorporates an IOM inlet and filter to gravimetrically collect the inhalable fraction of airborne dust. This can give a concurrent reference measure of inhalable airborne dust concentration. However, poor sealing within the sampling head resulted in some of the sampled dust not reaching the backup filter. This resulted in the Split 2 underestimating the reference inhalable dust concentration, which meant that it could not be accurately used as a calibration standard. Communications with the manufacturers have since revealed that the sampling head has recently been redesigned in order to improve the seal and eliminate leakage. The Respicon sampler gravimetrically underestimated the inhalable dust concentration, and did so increasingly as the particle size increased.

Keywords: calibration factor; dust monitor; inhalable dust; particle size

INTRODUCTION

There are increasing numbers of direct-reading dust monitors on the market today that will when properly calibrated give an accurate measure of airborne respirable dust, i.e. that which enters the mouth and nose

and passes to the lower regions of the respiratory system. These are mainly of the light scattering type where a light source (usually produced by a laser or diode) is collimated and illuminates dust entering the sensing volume. The intensity of the light scattered at a particular angle is proportional to the dust concentration. The monitors are usually calibrated in the factory using a 'standard' test dust and are adjusted to agree with respirable dust

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concentration measurements made using reference methods (HSL; MDHS 14/3, 2000). In reality they can be exposed to a wide range of dusts with differing physical properties such as particle size, refractive index and particle shape, which will affect their response by varying degrees.

Direct-reading dust monitors find uses in many different applications such as walk-through surveys, background sampling, site dust measurements, assessment of the effectiveness of dust control systems and measurement of indoor air quality (IAQ). They are also extensively used as part of an exposure video visualization system (Rosén *et al.*, 2005), to identify high levels of dust generated by poor work practice, in the investigation of dust control techniques and to generate hygiene training information. This system superimposes dust concentrations onto a video image of the work activity being carried out. This is currently being used to examine worker exposure to the respirable fraction of airborne dust. However, larger dust particles that deposit in the upper respiratory tract and form a major component of the inhalable fraction have been shown to be the cause of serious health problems. For example, workers exposed to sustained high levels of hard wood dust have been found to develop nasal cancer. Although there are gravimetric samplers available such as the IOM sampler, that will accurately measure personal exposure to inhalable dust concentrations, these will not measure short-term exposure levels and cannot be used with the aforementioned visualization system. There is therefore a requirement for a dust monitor that will accurately record and display the inhalable fraction of airborne dust in real time.

There are several direct-reading dust monitors that purport to measure or could be calibrated to measure mass concentrations of inhalable dust. However, these are all light scattering instruments, which scattering theory predicts are sensitive to dust particles primarily in the respirable size range. Therefore, they all need to be compared with a reference method in order to give an accurate measure of inhalable dust mass concentration. In some instances this is built into the instrument, whereas in others separate standard inhalable samplers such as the IOM sampler are required for comparison.

MATERIALS AND METHODS

Monitors tested

SKC Split 2. This measures the intensity of light scattered at 90° to determine the dust concentration. According to the manufacturer, it can be used for personal or area monitoring to measure respirable, thoracic or inhalable dust. It can be operated passively so that natural movement in the surrounding air introduces particles into the sensing chamber. It can also

be made active by combining the sampling head with a sample pump and cyclone (for respirable) or IOM inlet (for inhalable) for dust monitoring and concurrent dust sampling. The IOM inlet can also be fitted with calibrated foam plugs to selectively sample the respirable or thoracic fraction of the dust. A cartridge containing a 25 mm filter is fitted in the rear of the sampling head to capture the particles passing through the measuring zone and is used to determine the reference dust concentration.

The manufacturers claim a measuring range of 0.01–200 mg m⁻³ for particles of aerodynamic diameters from 0.1 to 100 µm. It will display instantaneous concentration in addition to TWA, Max, Min, STEL and time on a large LCD display. The data logging period can be set to 1 s, 1 min or 10 min and up to 21 500 data points can be stored. The logged data are downloaded to a personal computer using proprietary software included with the instrument. In the active configuration, the Split 2 is placed on one side of a worker's belt and the pump on the other. The sensor is mounted near the breathing zone for accurate breathing zone measurement.

To zero the Split 2 monitor, air is first drawn through an absolute filter placed on the IOM inlet using a small personal sampling pump. This ensures that the sensing zone is free of particles. It is then set to zero concentration according to the manufacturer's instructions. The Split 2 is calibrated by the manufacturer using SAEJ726 Fine Standard test dust (ISO 12103 Pt 1, 1997). This is largely composed of silica, and has a size distribution similar to Arizona road dust—another commonly used calibration dust. It is also supplied with a reference calibration element to carry out a single point check of the factory calibration.

Thermo Electron DataRAM (model pDR-1000). This is a small, lightweight, personal, forward light scattering (50–90° scatter angle) particle detection system, which relies mainly on ambient air movement rather than a pump to introduce particles into the sensing chamber. Larger particles, emitted as projectiles from energetic processes such as grinding or fettling can also enter the instrument under their own momentum. The sensing chamber is an integral part of the instrument, which is located underneath a rubber-protected hood through which the dust passes. It gives an optimum sensitivity to particles within the respirable range, determined by the variation of scattered light with particle size, which the manufacturers claim give it a high correlation with standard gravimetric measurements of the respirable fraction. The monitor is auto-ranging and covers a measurement range of 0.001–400 mg m⁻³. An RS232 digital serial connector is included so that readings can be displayed and recorded on a computer in real-time. A PC is used to program the operating and logging parameters via the serial port using the proprietary

software included. The monitor incorporates a data logger and the logged data can be downloaded to a PC. The data logging averaging period can be set between 1 s and 4 h. The monitor requires periodic zeroing, which is carried out by placing it inside a sealable polythene bag. A hand pump with an in-line absolute filter is used to partially inflate the bag with clean, particle-free air. The zero is then set on the instrument according to the manufacturers instructions. Unlike the Split 2 there is no physical method of adjusting the zero or checking the calibration. Like the Split 2, the DataRAM is also calibrated by the manufacturer to measure J726SAE Fine Standard test dust.

Respicon TM. The Respicon TM particle sampler is a lightweight, compact device for measuring personal exposure to airborne particle concentrations. It uses a single sampling head to model the human respiratory tract and simultaneously determine the three most critical particle fractions: inhalable, thoracic and respirable. The sampling head comprises a multistage, virtual impactor that traps airborne particles onto three individual collection filters. Airborne particles corresponding to the inhalable fraction are drawn into the Respicon through a ring-gap sampling inlet using a conventional personal sampling pump. Coarse particles pass straight through to the lower collector while other particles are aerodynamically separated onto the appropriate filter. The first virtual impactor stage separates out and collects particles smaller than 4 μm . The second stage collects particles below 10 μm , while the third stage collects the remaining coarse particles. The inherent design advantages of virtual impactation allow the Respicon to avoid many of the sampling problems common to conventional cascade impactors. Extended sampling periods are possible without the concern of overloading. Particle bounce losses are eliminated and there is no need for messy grease traps.

The TM model combines all the features of the standard Respicon sampler plus it also offers time-resolved concentration measurements of the three size fractions. The instrument combines both inertial classification with 90° light scattering photometric aerosol detection. The optical sensors are calibrated using the mass concentrations measured gravimetrically from the filter samples. An inherent decrease in mass-based photometer response with increasing particle size is partially offset by an increase in concentration of the coarse dust fraction. This is achieved because by design the larger particles are incorporated in a volumetric flow that is small compared to the total intake flow rate. The concentration is therefore enhanced by the ratio of the total flow rate to the minor flow rate, resulting in an increased sensitivity to larger particle sizes.

The TM model also incorporates a separate eight-channel data logger, three of which are used by the

Respicon to monitor the voltage output from the three photometers. The data logger can store up to 512 000 data records. The operating and logging parameters are input into the data logger module. The instrument is zeroed by operating it in a clean environment and then following the manufacturer's instructions. Data from the logger are downloaded and analysed using proprietary software supplied with the instrument.

Sibata PDS-2. Like the Dataram, this determines airborne dust concentration by measuring the intensity of scattered light in the forward direction. It is also most sensitive to particles within the respirable size range and its response is dependant on the physical and optical properties of the dust it is measuring. The PDS-2 introduces the dust actively through an annular inlet using a small fan incorporated into the sensing head. This dispenses with a separate sampling pump, making the instrument lighter. However, the fan is not powerful enough to draw the air through a filter and so a concurrent gravimetric sample is not possible. Therefore, a separate gravimetric sampler is required for calibration purposes.

The PDS-2 can be set to manual or auto-ranging and covers a measurement range of 0.001–100 mg m^{-3} . It has a built-in data logger with a capacity to store up to 60 000 measurements and the logging interval can be set between 1 and 60 s. Recorded data are downloaded to a PC via a built-in RS232 serial data port using a communications program such as the hyper-terminal program featured in Microsoft Windows. At the time of testing, the instrument was not supplied with proprietary software to do this. According to the instructions, the PDS-2 is calibrated in the factory using 0.6 μm latex particles. The instrument also comes supplied with a combined zero/calibration element. This replaces the sampling inlet and depending on the orientation can be used to check the zero or carry out a single point calibration check.

TSI Sidepak. This incorporates a laser photometer that measures the intensity of light scattered at 90° to determine the airborne dust concentration. It is a small, quiet, compact and lightweight aerosol monitor making it ideal for personal exposure monitoring. It can be worn comfortably on the belt to provide a continuous display of the real-time aerosol concentration for dust, fumes and mists. Aerosol is introduced into the monitor by a pump via an inlet port on the front of the instrument. The Sidepak covers a measurement range of 0.001–20 mg m^{-3} in the particle size range of 0.1–10 μm making it best suited to measurements of respirable dust concentrations. The integrated pump allows the use of size-selective aerosol inlet conditioners such as cyclones and impactors with 1, 2.5 and 10 μm cut-off points making it adaptable for both workplace and environmental airborne dust measurement. The pump is flow-controlled and is powered by a smart NiMH battery management

system that provides precise run time and battery condition information. The Sidepak displays both instantaneous and 8 h TWA dust concentrations. It also provides data logging of max, min, average, elapsed time and TWA concentrations and will record up to 31 000 readings. The logging interval is user adjustable between 1 s and 1 h. Data can be downloaded to a PC via a USB interface using TSI's 'Trakpro' proprietary software. The instrument is zeroed by attaching an absolute filter cartridge to the inlet and then following the manufacturers instructions. Calibration is carried out in the factory using SAEJ726 Fine Standard test dust. The monitor is not supplied with a calibration element with which to carry out a single point calibration check.

Although designed to measure respirable dust concentrations, with the application of a correction factor, measurements of inhalable dust concentrations might be feasible. The upper limit of 20 mg m^{-3} respirable will correspond to higher concentrations of inhalable dust, depending on the size distribution of the dust. With this in mind, an IOM inlet adaptor was made for the instrument. This consisted of a standard IOM inlet attached to the end of a pipe that had been bent through 90° . This was connected to the inlet nozzle on the Sidepak using a short length of Tygon tubing. Particle losses into the inlet were minimized by optimizing the pipe diameter, bend radius and tube length using theoretical predictions summarized by J. E. Brockman (Willeke and Baron, 1993). These show that for the pipe used (internal diameter 1.4 cm and bend radius 5 cm) particle losses will be relatively low for particle sizes up to $30 \mu\text{m}$ (<30% for $30 \mu\text{m}$ particles) but will increase significantly for particles larger than this size.

Test dust samples

Stone dust, wood dust (sieved white pine) and coal dust were selected to represent dusts found in typical industries such as building and construction, wood-working and coal mining. The dusts were also chosen to have different physical characteristics such as refractive index, density and size distribution. Three grades of aluminium oxide [trade name Aloxite—supplied by Washington Mills (UK) Ltd] were chosen to investigate the effects of particle size on instrument response. These were 1200, 800 and 320 grades which represent quoted mass median aerodynamic diameters (MMAD) of 6, 13 and $58 \mu\text{m}$, respectively. Aloxite is an abrasive dust that finds uses mainly in grinding applications but an additional major advantage is the narrow distribution of the particle size for each grade, making it ideal for investigating the effects of particle size on the performance of aerosol measuring devices (Mark *et al.*, 1985). The particle size data were provided by the manufacturer, but were also checked using methods described later.

Realstone Ltd (UK) provided the stone dust: stone was cut using saws and the dust produced was extracted at source, filtered and collected inside a bag. The dust was slightly agglomerated due to the damp conditions and was therefore dried inside an oven and sieved prior to use. The pine dust was bought pre-sieved (WTL International Ltd, UK) and was graded as 180 (sieve aperture, 0.09 mm). The coal dust was Silkstone coal dust, which is coal that has been taken from the 'Silkstone seam' at Selby colliery. This is then ground and sieved at the Health and Safety Laboratory to give particles no larger than $200 \mu\text{m}$.

Methodology

In order to determine the suitability of the direct-reading dust monitors for measuring airborne inhalable dust they were compared with the IOM personal inhalable sampler (Mark and Vincent, 1986).

Although the main aim of this study was to measure the inhalable fraction of airborne dust, additional standard respirable samplers were also included in the tests. The conducting plastic version of the Casella respirable cyclone sampler was used as the reference sampler for comparison with responses of the direct-reading instruments.

The tests were carried out inside a large recirculating wind tunnel (Blackford and Heighington, 1986), which provided a means of exposing dust monitors and reference dust samplers to dust concentrations under controlled steady state conditions. With the correct choice of dust generator and dust injection system, constant and uniform dust concentrations and velocities were achieved over an extended period of time. The samplers and monitors were mounted onto a metal manikin placed inside the tunnel, to ensure that the samplers collected the correct size fraction of the dust.

Simplified metal manikin—the CALTOOL

The CALTOOL (short for calibration tool) is a reference instrument designed for real workplace aerosol measurements that is used to provide a standard against which candidate samplers can be compared (Kromhout *et al.*, 2006). It comprises a cylindrical stainless steel head with circular mouth entry through which air is sampled at 20 l min^{-1} . The aerosol contained in the sampled air is collected in a 47 mm filter holder. The head is mounted onto a stainless steel upper torso of elliptical cross-section upon which candidate samplers are mounted. The torso is normally connected via a telescopic pole to a wheeled pumping box, containing a large pump that provides suction for both the CALTOOL sampling head and up to six personal samplers. For the purpose of these tests the pumping box was removed and the telescopic pillar was mounted onto a rotating

turntable. This turned at 0.5 r.p.m. and had a reciprocating action so that any sampling biases caused by the relative motion of dust samplers to air flow were minimized. Also, the reciprocating action meant that the sampling pumps for the reference samplers and dust monitors could be placed outside the tunnel and the connecting air tube would not become twisted. This had the added advantage of preventing the pumps becoming contaminated by the dust being generated.

The samplers and dust monitors to be tested were mounted on the front and rear of CALTOOL using hook and loop tape for rapid attachment and release.

Large recirculating wind tunnel

A detailed description of the tunnel is given by Blackford and Heighington (1986), although subsequent modifications to the dust generation and injection system have been made and will be described later. Turbulence grids were also inserted to allow better mixing of the injected dust. Figure 1 shows a schematic of the tunnel. Since several samplers and dust monitors were exposed simultaneously during a test, it was important that measurements were not influenced by concentration or velocity gradients inside the tunnel, especially in the central area where the manikin was situated. Although these have been checked for a previous project (Thorpe and Walsh, 2001), the tunnel layout, sampler locations and dust injection system have all changed since then. Therefore, preliminary measurements of dust concentration and air velocity uniformity were made at the position where the tests were carried out.

Velocity profile. This was determined by traversing the tunnel cross-section using a calibrated hot wire anemometer (Dantec Electronics Ltd). Exact positioning of the probe was achieved using a grid made from 3 mm diameter galvanized steel with a cell size of 75×75 mm, which covered the entire

cross-section of the tunnel. The anemometer probe was placed at the centre of alternate cells and the velocity was measured. A calibrated reference anemometer (TSI Inc.) was also placed centrally inside the tunnel just downstream of the grid so that the velocity profile could be corrected for any temporal changes in velocity, although in practice the velocity inside the tunnel was very stable. A profile was produced for a nominal velocity of 0.5 m s^{-1} . The results showed good uniformity over the entire tunnel cross-section with a coefficient of variation (CV) of 7% from a mean value of 0.57 m s^{-1} . The uniformity was slightly better over the area that the manikin occupied with a CV of 6% from a mean value of 0.56 m s^{-1} . The profile was carried out using the air-operated dust injectors working, but not generating dust. Although the injector configuration (spacing, distance from samplers and operating air pressure) was sometimes changed from test to test depending on the type of dust being generated, periodic checks on the velocity profile showed very little change.

It should be noted that the introduction of a blunt body such as a manikin will affect the uniformity of air flow within the tunnel depending on the size of the blockage. The blockage of the tunnel cross-section provided by CALTOOL was $\sim 13\%$.

A new dust generation system

In order to investigate particle size effects the dust was injected as close as possible to the position of the sampling instruments, whilst also maintaining a uniform velocity and dust concentration profile over the sampling area.

A multipoint injection system was used to achieve this, which was similar to that used by Heist *et al.* (2003). However, rather than sucking the dust from a moving conveyor belt simultaneously into an array of air-aspirated nozzles, the dust was fed into a row of dust injectors using a scanning screw feed system as

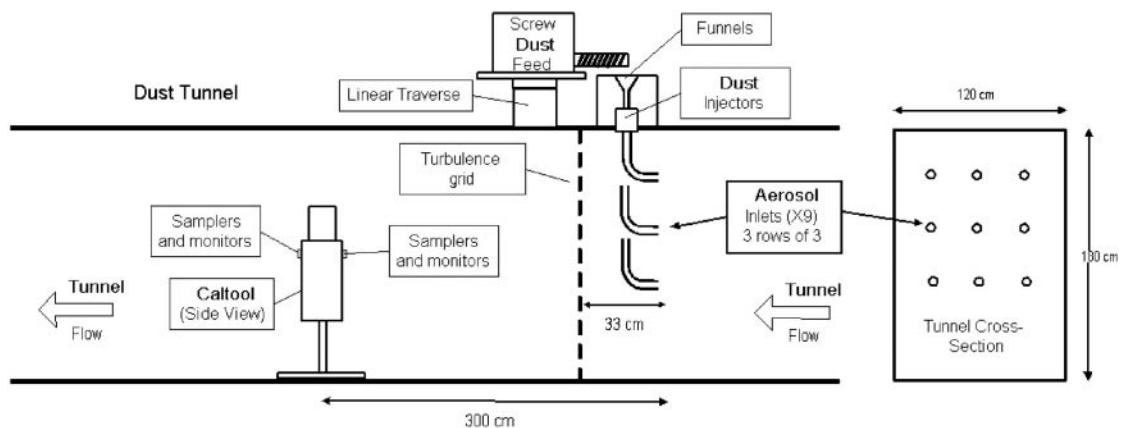


Fig. 1. Experimental set-up inside the large dust tunnel.

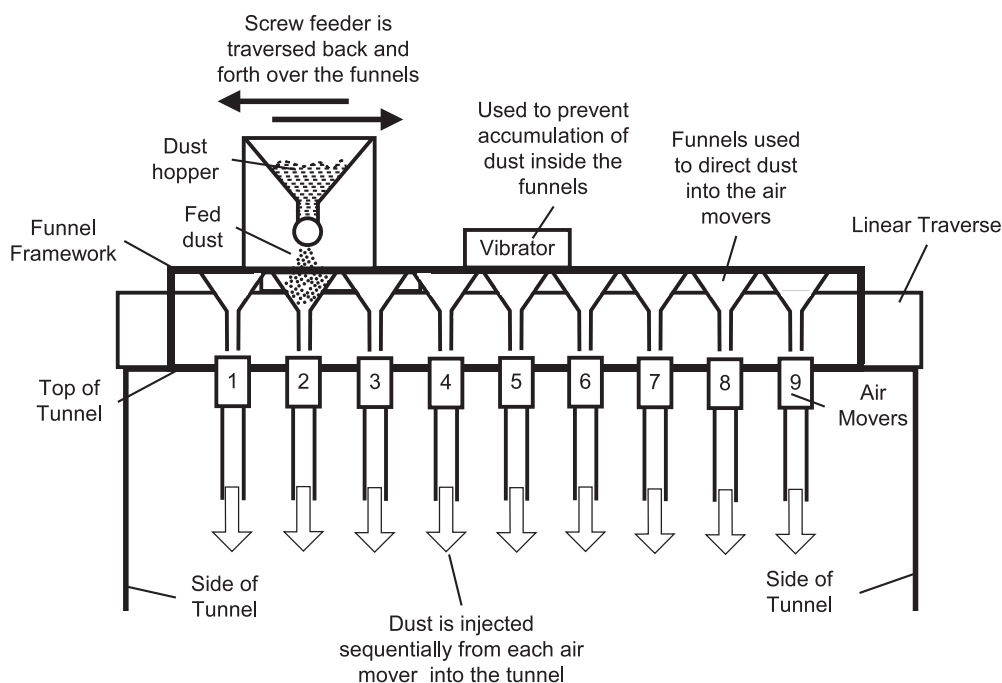


Fig. 2. Schematic of the dust feed system.

illustrated in Fig. 2. This was achieved by placing commercially available air movers through the top of the tunnel and across its width. Regulated compressed air lines were used to power these individually. The screw dust feeder was traversed slowly back and forth across the tunnel using a linear module with speed control, which was also attached to the top of the tunnel. The dust from the rotating screw fell onto a row of funnels, directly below each was one of the air movers. The dust was then injected from the air movers into the tunnel through an array of nine 22 mm diameter copper pipes, which were directed into the tunnel airflow. From here the dust passed through a turbulence grid to reduce any large-scale turbulence and then on to the sampling zone. More details of the method are given in a separate paper (A. Thorpe, submitted for publication).

Because the dust is fed sequentially into the funnels, the dust concentration inside the tunnel at any one point in time will be non-uniform. In order to overcome this problem, tests were carried out over extended periods so that the dust feed traversed the tunnel section many times resulting in each air mover receiving the same average amount of dust.

Concentration profile measurement. For the purpose of these tests a concentration profile measurement over the complete cross-section of the tunnel was not required. Dust uniformity was only required over the area where the reference samplers and dust monitors were placed. The largest size of Aloxit dust ($d_{ae} = 58 \mu\text{m}$) was used, as this was assumed to be the most problematic to generate uniformly owing to its

high settling velocity. The dust injector configuration and tunnel set-up was that shown in Fig. 1. Six IOM samplers fitted with 25 mm glass fibre filters were placed on the upper torso of the metal manikin (three on the front and three on the back) in the breathing zone. The dusty air was drawn through the samplers at 2 l min^{-1} using small, flow-regulated personal sampling pumps. The samplers were run for long enough to collect a weighable ($>500 \mu\text{g}$) dust sample on the filters. The resultant variation in concentration for the six samplers was $\pm 8\%$. This was regarded as acceptable and it was assumed that the variation would be as good if not better for the smaller particle sizes.

Tunnel experiments

The reference samplers and dust monitors were arranged on the manikin as shown in Fig. 3 at the position inside the tunnel shown in Fig. 1. They were placed either side of the manikin torso so that they could be better spaced. One reference personal inhalable dust sampler (IOM) and one reference personal cyclone respirable sampler (SIMPEDS) was placed either side of the manikin. An 8-stage series 290 Marple personal cascade impactor (Graseby Andersen Ltd) was also included for at least one test at each particle size. This gives a measurement of the aerodynamic particle size distribution for the dust being sampled. Adjusting the sample flow rate (between the limits $0.5\text{--}5 \text{ l min}^{-1}$) into the impactor varies the size range of the particles sampled. The flow rate was nominally set at 2 l min^{-1} to give a

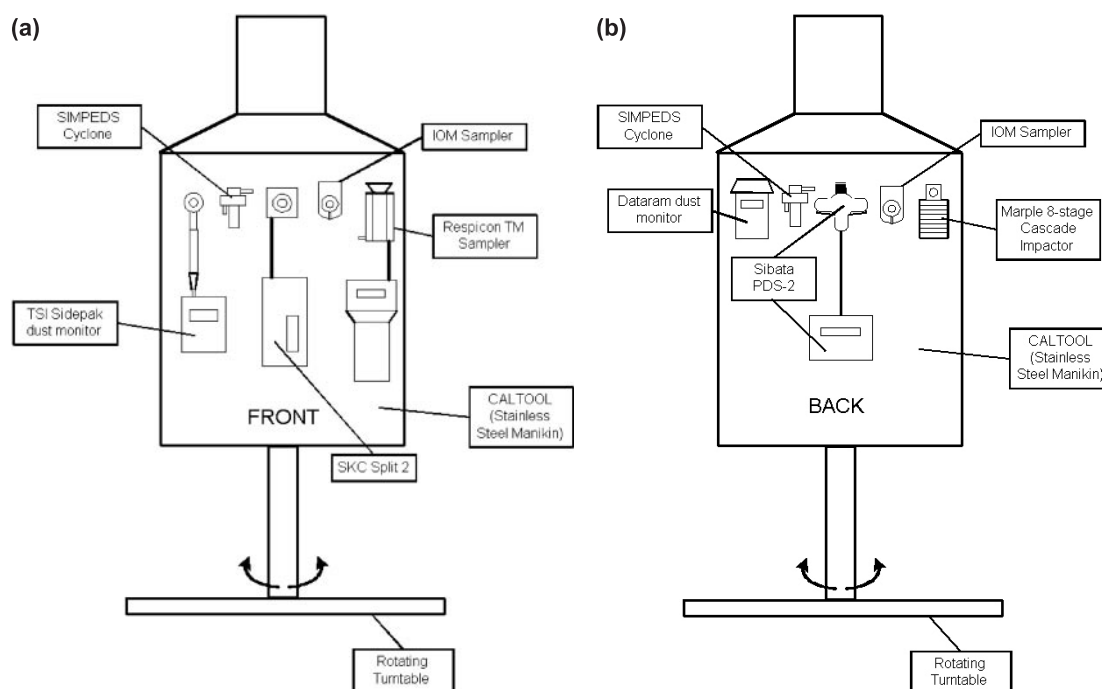


Fig. 3. (a) Monitors placed on the front of the CALTOOL. (b) Monitors placed on the rear of the CALTOOL.

Table 1. Measurement of particle size of test dusts using the Coulter Multisizer and the Cascade Impactor

Dust type	Nominal		Coulter Multisizer			Impactor MMAD (μm)
	MMAD (μm)	σ_g	Volume median (μm)	Number median (μm)	σ_g	
Aloxite 240	89.5	1.29	80.6	24.4	2.56	
Aloxite 320	58	1.17	63.1	31.4	1.99	
Aloxite 400	34	1.2	34	29.7	1.22	
Aloxite 800	13	1.3	17.5	10.5	1.45	19.9
Aloxite 1200	6	1.36	7.7	4.8	1.54	7.3
Stone dust			20.3	3.1	1.75	21.6
Coal dust			21.1	2.9	1.96	22.7
Pine dust			19.8	6	2	20.4

σ_g = the geometric standard deviation.

measurement range of 0.5–21 μm , although this was reduced for some of the larger dusts to increase the upper size limit. Before every test, each impactor stage was fitted with a new Mylar collection substrate onto which the dust impacts and collects. These were coated, using a small artist's brush, with a 20% solution of high vacuum grease dissolved in toluene. This was to encourage the larger particles (those with the highest momentum) to adhere to the surface so that they did not bounce off and pass on to the next impactor stage. To further prevent particle bounce, the experiments were designed to ensure that the substrates were not overloaded with dust, which would reduce the adhesive properties of the greased surface. In operation, the largest particles are impacted and collected onto the first stage with progressively smaller particles being inertially separated and collected

onto the later stages. The coated substrates were weighed before and after the test.

There is a range of views as to the best way of preventing particle bounce. Our experience is that vacuum grease dissolved in toluene gives satisfactory results, and this was confirmed in this experiment by the good agreement of the cascade impactor with other methods, as reported below. We also strove to ensure that the loading of the Mylar substrate was kept to a minimum to reduce any effect.

At the start of testing only the Sibata PDS-2 was new and unused. The other monitors had been used to various degrees, but they had all been regularly serviced and calibrated when required and were all operating correctly at the time of testing. The cyclones, IOM samplers (including the Split 2 IOM inlet) and Respicon were loaded with clean glass fibre filters,

Table 2. Calibration factors for all monitors and all dust types

Dust Type	Mass median aerodynamic particle diameter (Impactor) (μm)	Mass median aerodynamic particle diameter (Coulter) (μm)	Dust monitor	Calibration factors measured as the inverse slope of the graph (IOM/monitor) for inhalable results	Variance (R^2)	No. of readings	Calibration factors measured as the inverse slope of the graph (cyclone/monitor) for respirable results	Variance (R^2)	No. of readings
Aloxite (1200 grade)	7.3	7.7	Split 2	5.03	0.9976	5	0.504	0.9917	5
			Dataram	6.02	0.9998	5	0.604	0.9856	5
			Sidepak	6.84	0.986	3	1.044	0.9872	3
			Sibata PDS-2	18.48	0.9975	5	1.853	0.991	5
Aloxite (800 grade)	19.9	17.5	Split 2	17.64	0.9684	6			
			Dataram	26.67	0.9903	6			
			Sidepak	24.39	0.9827	6			
			Sibata PDS-2	128.21	0.9562	5			
Aloxite (320 grade)		63.1	Split 2	181.82	0.8953	4			
			Dataram	714.29	0.914	4			
			Sidepak	588.24	0.9777	4			
			Sibata PDS-2						
Pine dust	20.4	19.8	Split 2	11.83	0.981	5			
			Dataram	18.73	0.9838	5			
			Sidepak	17.45	0.9979	3			
			Sibata PDS-2	62.11	0.9811	5			
Stone dust	21.6	20.3	Split 2	10.26	0.9842	5	0.569	0.8905	4
			Dataram	11.33	0.989	5	0.626	0.9556	4
			Sidepak	8.85	0.9838	3			2
			Sibata PDS-2	27.47	0.9976	5	1.526	0.884	4
Coal dust	22.7	21.1	Split 2	12.61	0.9927	4	1.126	0.9517	4
			Dataram	12.41	0.9994	5	1.113	0.9939	5
			Sidepak	11.90	0.98	5	1.070	0.962	5
			Sibata PDS-2	35.84	0.9518	4	3.215	0.9436	4

which were conditioned and weighed before and after exposure. Air was drawn through these at flow rates of 2.2, 2 and 3.11 l min^{-1} , respectively, using personal self-regulating sampling pumps. The flow was checked before and after each test using a calibrated bubble flow meter with a range of $0\text{--}6 \text{ l min}^{-1}$ (Gilian Ltd). The flow rates into each stage of the Respicon sampler are also critical and if adjusted correctly will ensure that it correctly samples each particle size fraction. Therefore, the flow rates were checked periodically using a flow calibrator that is fitted in place of the filter stages. The measured flow rates were always within the manufacturer's quoted tolerances. Each dust monitor was zeroed prior to each test, in accordance with the manufacturer's instructions and they were each programmed to record a measurement every 5 s. The default flow rate for the TSI Sidepak dust monitor is 1.7 l min^{-1} . During these tests the instrument was used with the purpose built IOM inlet as described previously. The flow rate was adjusted to 2 l min^{-1} so that the IOM inlet operated at its correct flow rate. The Split 2 and Sibata PDS-2 dust monitors were calibrated before each test using the supplied calibration element.

Tests were carried out at a tunnel air velocity of 0.5 m s^{-1} for each test dust, over a range of concentrations. The concentration was varied by either changing the speed of rotation on the feed system and/or by changing the size of the screw. The test duration was varied according to the dust concentration but was usually between 30 min and 2 h so that a weighable dust sample was obtained ($>500 \mu\text{g}$) on the filter.

Sizing of the airborne test dust

As mentioned previously, the aerodynamic size distribution of the dust was measured using a personal cascade impactor mounted on the manikin. These can suffer practical drawbacks that need to be considered. For instance, for some tests the quantity of dust collected on the Mylar substrates was small ($<100 \mu\text{g}$), usually when the concentration of dust was low and/or when the cascade impactor was operated at a low flow rate. This will result in greater errors in the determination of the MMAD of the aerosol. Also, particle bounce within the impactor can lead to sizing errors and so efforts as described earlier were made to reduce the effects.

Therefore, as a further measure of the particle size, samples from the bulk dusts were analysed and sized using the Coulter Multisizer. This uses the Coulter principle (electrical sensing zone method) to measure the particle volume. The Multisizer was calibrated prior to use with monodisperse latex spheres whose size is traceable to NIST standards. The measured equivalent volume diameters were converted into aerodynamic diameters by multiplying by the square

root of the specific gravity of the material being analysed.

Five samples of each dust suspension were analysed and an average of the readings were taken. The Multisizer can suffer from particle coincidence effects whereby more than one particle passes into the sensing zone at any one time. This occurs at high particle concentrations and so care was taken to keep the concentration below the coincidence limit by adequate dilution of the particle suspension. Also, care was taken to ensure that the tube orifice did not become blocked during analysis.

RESULTS AND DISCUSSION

Particle size measurement

The results of the particle sizing measurements are summarized in Table 1. The cascade impactor data were reduced using algorithms described in the instruction manual. These were used to produce a computer spreadsheet into which the filter weights could be entered and which were then used to calculate and plot the particle mass size distribution. The MMAD is defined as the diameter at which 50% of the particle mass is less than the indicated size. Curve fitting software (Sigmaplot, Jandel Ltd) was used to fit the best curve to the data, which was usually a sigmoid fit. The MMAD was then calculated from the resultant equation of the fitted curve. The geometric standard deviation (σ_g) has also been calculated for

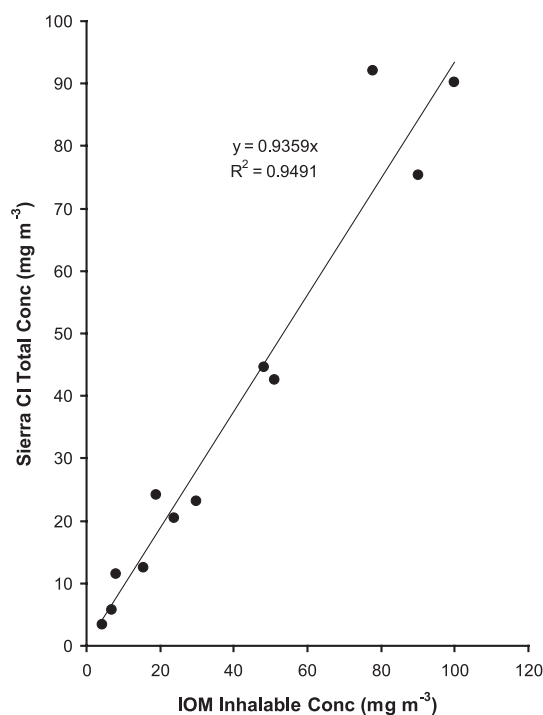


Fig. 4. IOM inhalable concentration versus Sierra Cascade Impactor total concentration for all dusts.

each dust from the Multisizer results and they are shown in Table 2. These were determined from the ratio of the 84th percentile to the 50th percentile of the size distribution. The geometric standard deviation

gives an indication of the width of a size distribution (or degree of polydispersity). The larger the value the greater is the spread in the size distribution. A perfectly monodisperse dust would have a $\sigma_g = 1$.

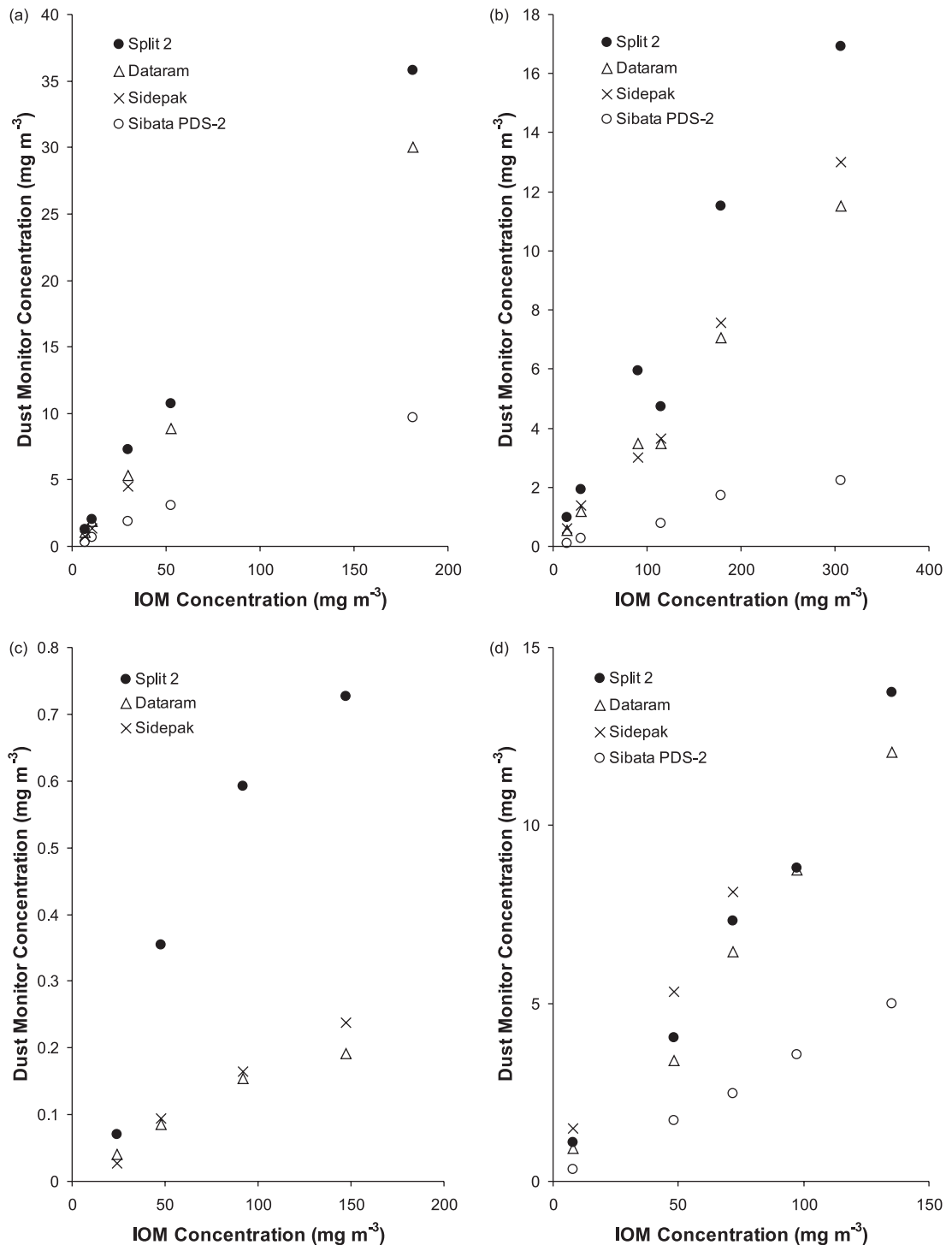


Fig. 5. (a) Monitor response versus IOM concentration for 1200 grade aloxite (6 μm). (b) Monitor response versus IOM concentration for 800 grade aloxite (13 μm). (c) Monitor response versus IOM concentration for 320 grade aloxite (58 μm). (d) Monitor response versus IOM concentration for stone dust. (e) Monitor response versus IOM concentration for pine dust. (f) Monitor response versus IOM concentration for coal dust.

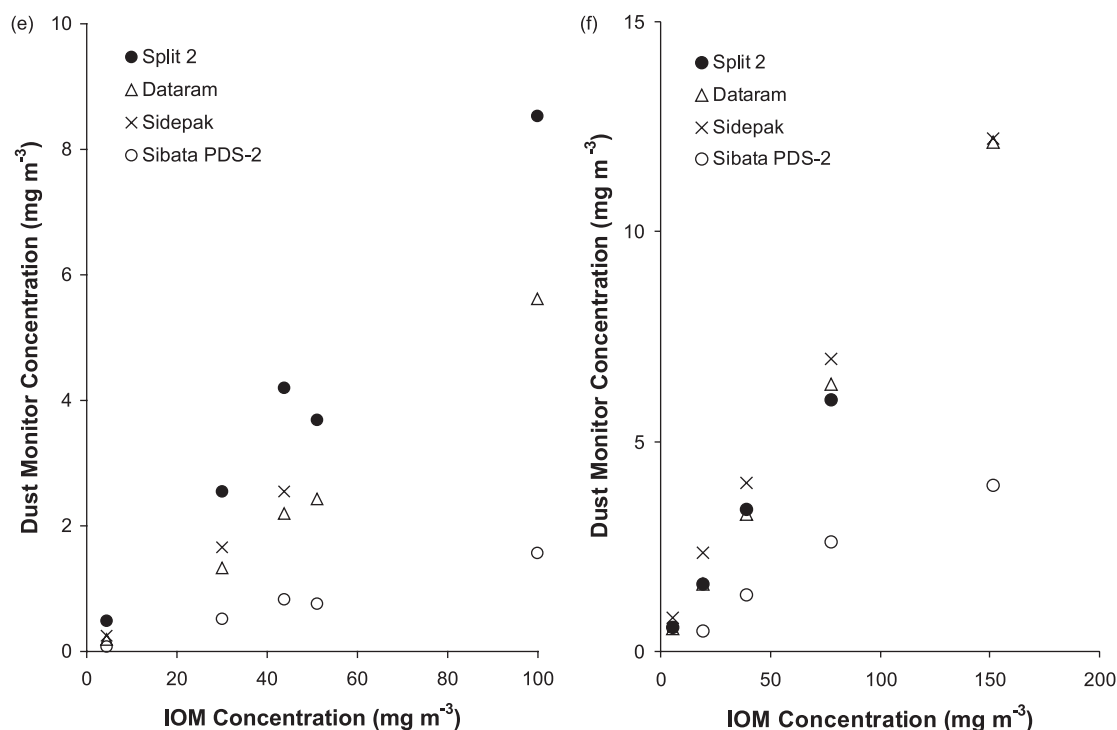


Fig. 5. Continued

Therefore, an aerosol with a large median diameter may still contain particles in the respirable size range if the σ_g is high.

Table 1 shows remarkably good agreement in the measurement of the MMAD for all dusts for both particle sizing techniques, although a greater range of sizes would have been preferred for comparison. The Multisizer also agrees closely with the nominal manufacturers' quoted sizes for the different grades of Aloxite dust. Such close agreement would appear to confirm the accuracy of both the cascade impactor and the Multisizer for determining the size distribution of the test dusts. The values of σ_g are fairly typical of many aerosols found in workplace situations.

The weights of dust deposited on each impactor stage can be summed to give a measure of the total airborne dust concentration. For the size of dusts used this should be close to the inhalable concentration. Figure 4 compares the total cascade impactor concentration with the measurement of inhalable dust concentration made with the IOM samplers. As can be seen there is a reasonable linear fit to the data and the relationship is close to one.

Monitor results

Calibration factors. The responses of the dust monitors to dusts of different size distributions and composition as functions of both inhalable and respirable dust concentrations are shown in Figs 5 and 6. Calibration factor is defined as the number that

the monitor concentration is multiplied by so that it agrees with the reference measurement. Mean calibration factors (the inverse slope of the graphs) for the different monitors are also summarized in Table 2.

Each monitor greatly underestimated the inhalable concentration for all the dusts tested. This was not surprising since all the dusts contained particles larger than respirable, i.e. within the only easily detected fraction. Also, their response decreased with increasing particle size as shown by the instrument response curve in Fig. 7, for the three grades of Aloxite. The instrument response is the inverse of the calibration factor. The reason for the decrease is because the proportion of respirable dust drops rapidly with increasing particle size. The instrument response axis is plotted on a logarithmic scale because of the large variation in response with particle size.

All monitors showed good linearity when compared with the reference IOM inhalable concentration, for all dusts apart from the 320 grade Aloxite (MMAD = 58 μm). Monitor response to this dust was very low since it contains very few particles in the respirable size range. The response is non-linear and appears to fall with increasing concentration. This is probably because the background dust concentration is significant compared to the generated respirable dust concentration. As the Aloxite concentration is increased, the effect of the background dust levels would diminish resulting in the observed fall in response.

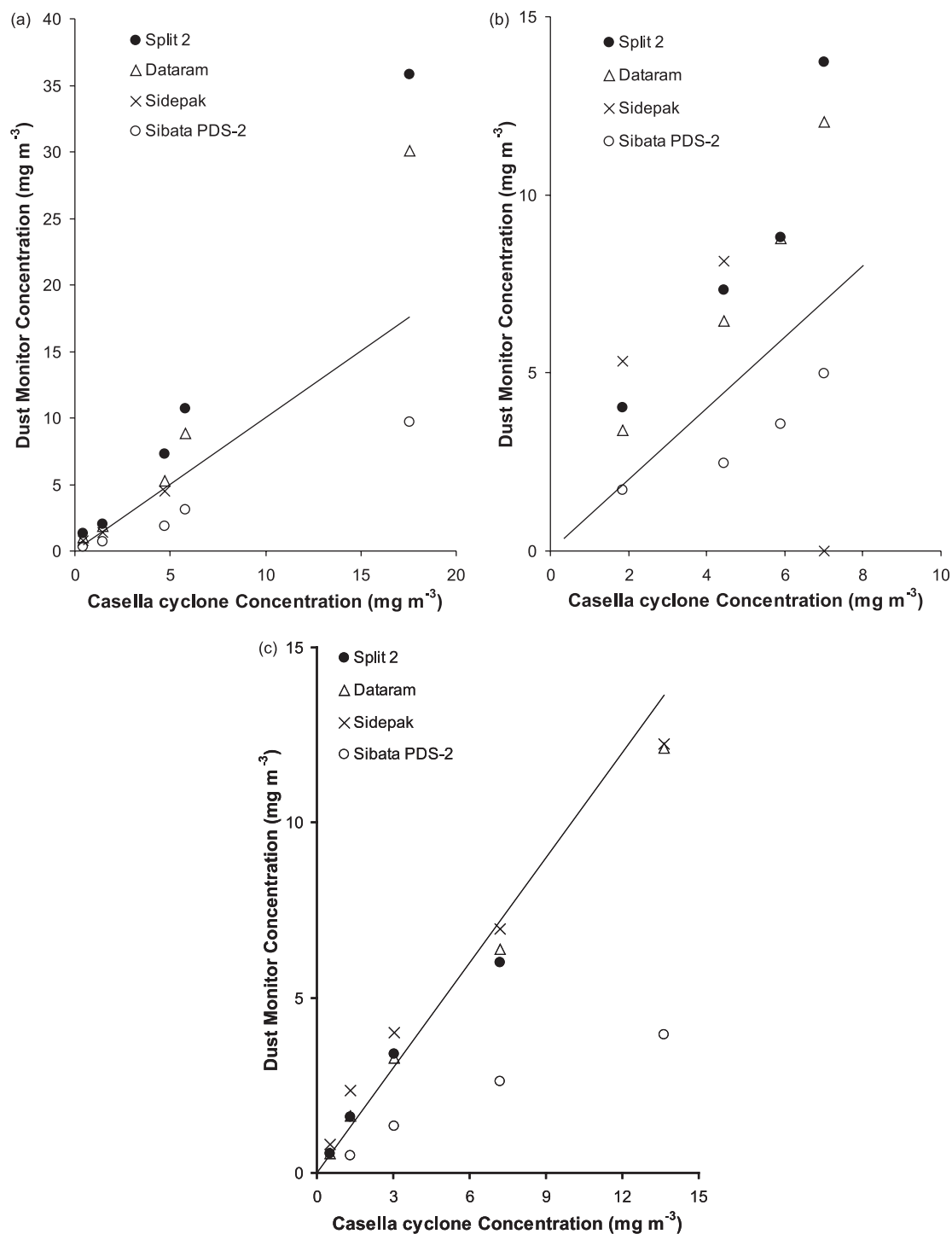


Fig. 6. (a) Monitor response versus cyclone concentration for 1200 grade Aloxit ($13 \mu\text{m}$). (b) Monitor response versus cyclone concentration for stone dust. (c) Monitor response versus cyclone concentration for coal dust.

The Dataram and Sidepak dust monitors showed very similar responses for all dusts as indicated by the average calibration factors shown in Table 2. The Split 2 generally gave a higher response (lower calibration factors) for all dusts and sizes, apart from

coal dust where it was about the same. The Sibata dust monitor always showed a lower response for all dusts and sizes resulting in higher calibration factors. This can probably be explained by the way in which the dust enters the instrument rather than its

light scattering characteristics, which should be similar to the other monitors. The dust enters the instrument through an inlet consisting of a series of annular disks mounted concentrically on top of each other, with a small gap between them. These disks could feasibly act as elutriators and/or impactors onto which the dust collects, thereby reducing the entry efficiency.

When monitor response was compared with the reference cyclone respirable dust concentrations there was much more scatter in the results, as shown in Fig. 6, and by the lower R^2 values shown

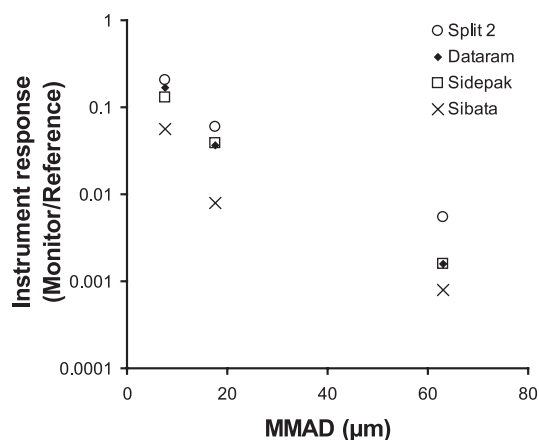


Fig. 7. Change in dust monitor response with particle size.

Table 3. Respicon inhalable photometer response as a function of particle size

Dust type	MMAD (μm)	Mean photometer response (ratio of inhalable photometer voltage to gravimetric measurement) (mV per mg m ⁻³)	SD	CV
Aloxite 320	63.1	1.42	Only two readings	
Aloxite 800	19.9	7.1	2.1	29.4
Aloxite 1200	7.3	11.3	2	11.3
Pine	20.4	11	0.37	3.3
Stone	21.6	8.8	0.94	8.8
Coal	22.7	2.2	0.17	7.8

Table 4. Ratio of Respicon inhalable to IOM inhalable concentration as a function of particle size

Dust type	Mass median aerodynamic particle diameter (impactor)	Mass Median aerodynamic particle diameter (coulter)	Mean ratio of Respicon inhalable to IOM inhalable concentrations	CV of mean ratio of Respicon inhalable to IOM inhalable concentrations (%)
Aloxite 320		63.1	0.7	9.7
Aloxite 800	19.9	17.5	0.86	12.2
Aloxite 1200	7.3	7.7	1.02	8.2
Pine	20.4	19.8	0.7	6
Stone	21.6	20.3	0.81	4.6
Coal	22.7	21.1	0.91	10

in Table 2. The main reason for this was the smaller quantities of dust collected on the filters compared to the IOM samplers. This was especially the case for the larger sizes of dust where most of the particles are beyond the respirable range and so will not be collected by the cyclone sampler. Because of this, there are only plots for the finer dusts, i.e. stone dust, coal dust and 1200 grade Aloxite. The weighing errors for the coarser pine dust and the 320 and 800 Aloxite grades were too great to give meaningful results.

The Split 2 and Dataram dust monitors generally overestimated the respirable dust concentration for Aloxite 1200 and stone dust, resulting in calibration factors less than unity as shown in Table 2. The reason for this is probably a combination of several factors. First, all dust monitors were calibrated using ISO 12103 Pt 1 dust whose light scattering properties are probably different from the dusts being tested.

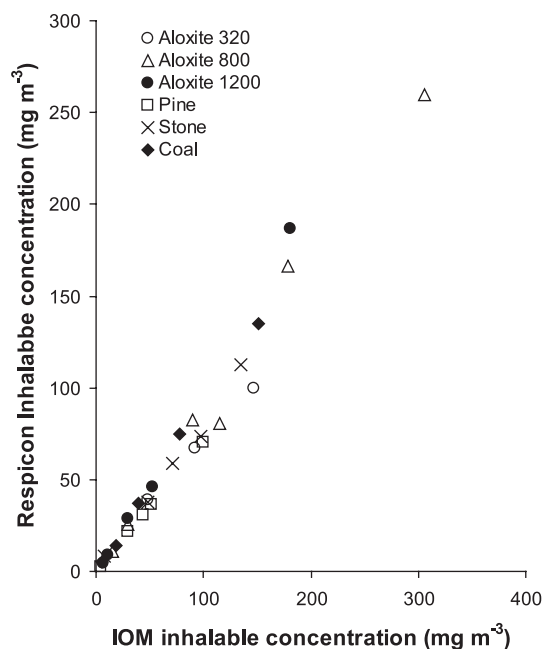


Fig. 8. Respicon inhalable concentration versus IOM inhalable concentration.

Second, the standard BS EN 481 (BSI, 1993) shows that when the dust particle size exceeds $\sim 12 \mu\text{m}$, respirable samplers (cyclones) cut-off and stop sampling, if they correctly follow the convention. However, light scattering monitors can continue to detect particles, to a certain extent, up to $\sim 30 \mu\text{m}$ in diameter. Therefore, as the particle size increases, light scattering monitors increasingly overestimate the true respirable dust concentration. The Sibata PDS-2 repeatedly underestimated the respirable concentration resulting in calibration factors greater than unity. This was once again probably owing to the inlet design rather than light scattering effects. The Split 2 and Dataram monitors agreed closely with the reference respirable concentration when used to measure coal dust, indicated by calibration factors close to unity in Table 2.

Respicon TM. The Respicon is considered separately because, although it uses light scattering to measure dust concentration, it does not display this in real-time on the instrument. Instead the dust concentration of each size fraction is determined after a test using the weight of dust collected on each impactor stage. The logged voltages for each impactor stage are downloaded to a PC using proprietary software supplied with the instrument. The software then applies the average values of the photometer voltage measured at each impactor stage to the gravimetric measurements to give a photometer calibration. This calibration is then applied to each data point to give a graph of dust concentration against time. The calibration factors for each photometer can be saved to PC so that if the Respicon is used to monitor the same dust at a later time, then gravimetric measurements will not be required. Instead, the calibration factor can simply be applied to the photometer voltages. However, care should be taken when doing this, since any changes in the physical properties of the dust will significantly affect the photometer voltage output as shown in Table 3.

The accuracy of the Respicon relies on weighable quantities of dust being collected on the three filter stages. Very small dust samples were observed on the first two stages, especially at large particle sizes. This made determination of the respirable and thoracic dust concentrations very inaccurate for most of the tests. Therefore, only the inhalable concentration was considered. The results for the Respicon inhalable dust concentration measurements when compared with the reference IOM sampler are shown in Fig. 8 and Table 4. The low values of the CV shown in Table 4 show that the relationship is very linear. Also, it can be clearly seen that as the particle size increases the Respicon progressively gravimetrically underestimates the inhalable concentration (shown best by the Aloxite results). At the maximum particle

size of $63 \mu\text{m}$ MMAD the Respicon gravimetrically underestimates the inhalable concentration by $\sim 30\%$.

Split 2 dust losses. Table 5 compares gravimetric measurements of the inhalable dust concentration made with the reference IOM sampler with those made with the IOM sampler fitted to the Split 2 dust monitor. As can be seen, the measurements made with the Split 2 monitor are considerably lower than those made with the reference IOM sampler. On inspection of the Split 2 sampling head there was evidence of dust leaking around the filter cassette as shown in Fig. 9. This shows that a poor seal existed between the filter cassette and the rear of the sampling head. In an attempt to improve the seal and eliminate leakage, a rubber gasket was placed between the rear of the sampling head and the filter cassette. This resulted in lower losses as indicated in Table 5. However, the Split 2 still underestimated the gravimetric inhalable concentration by an average of 32% and there was still visible evidence of leakage. Discussions with the manufacturers have since revealed that the sampling head has now been redesigned to eliminate this problem, although we have not been able to test the modified design.

Table 5. Effects of poor seal of filter cassette inside the Split 2 dust monitor

Test dust	Gravimetric inhalable dust concentration		Split 2 percentage loss of dust
	Reference IOM	Split 2	
Aloxite 320	92.03	44	52.2
Aloxite 320	147.14	45.55	69.0
Aloxite 800	90.03	34.34	61.9
Aloxite 800	114.62	49.62	56.7
Aloxite 800	29.66	13.57	54.2
Aloxite 800	305.86	140.5	54.1
Aloxite 800	178.87	101.04	43.5
Average percentage Loss = 55.9 (CV = 14.3)			
Improved seal on Split 2 filter cassette			
Aloxite 1200	181.27	155.35	14.3
Pine	43.74	26.14	40.2
Pine	43.74	26.14	40.2
Pine	30.02	18.83	37.3
Pine	51.04	32.6	36.1
Stone	48.21	25.9	46.3
Stone	97.45	64.54	33.8
Stone	71.77	26.54	63.0
Stone	135.21	87.78	35.1
Coal	77.82	52.96	31.9
Coal	39.38	31.36	20.4
Coal	5.46	3.99	26.9
Coal	151.61	146.96	3.1
Average percentage loss = 32.4 (CV = 47.5)			

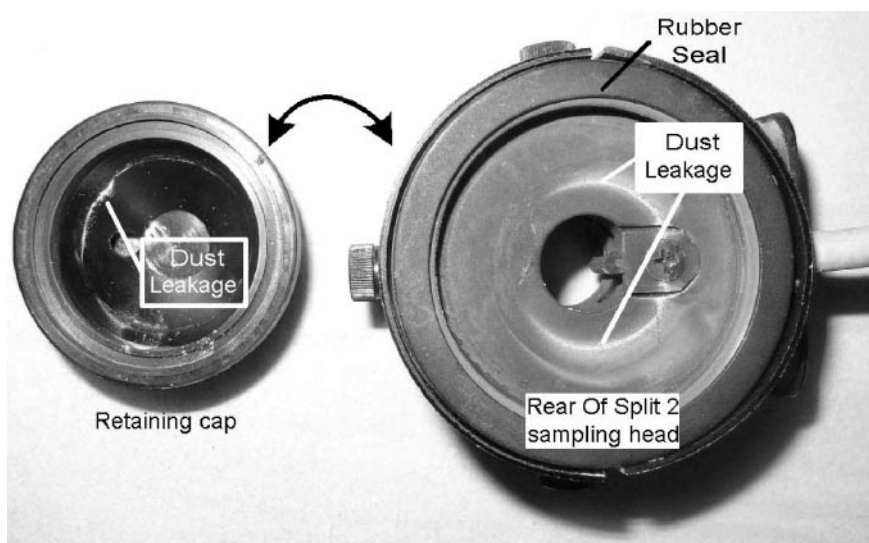


Fig. 9. Dust leakage inside Split 2 sampling head.

Cleaning the monitor optics. The dust monitors fall into two groups as regards cleaning the optics. First, there are the Split 2, Sibata PDS-2 and the Respicon TM monitors whose optics are not well shielded from the dust and so can become contaminated very quickly. They were regularly cleaned during the tests using a cotton swab soaked in alcohol which is not a difficult task since the optics are generally quite accessible. Second, there are the Dataram and Sidepak monitors, whose optics are well shielded from the dust and so become contaminated less frequently. However, these are very difficult to access and clean and so are best returned to the manufacturer.

CONCLUSIONS

A method for generating uniform concentrations of large particulate at low wind speeds inside a dust tunnel has been developed. This has enabled us to investigate the suitability of using direct-reading dust monitors for measuring the inhalable fraction of airborne dust.

The Split 2, Dataram, Sidepak and Sibata PDS-2 dust monitors all greatly underestimated the measurement of inhalable dust concentration. Their responses were found to be linear, but decreased rapidly as the particle size increased. Therefore, a calibration factor could be applied to the monitor readings to give an accurate measure of inhalable dust, but only if the particle size did not change. Ideally the instruments should always be operated alongside reference inhalable samplers if quantitative rather than qualitative measurements are required. For short-term measurements and/or measurements of low concentrations this might not be possible because the quantities of

dust collected on the filters would be too low to weigh.

There was much more scatter in the results when the dust monitors were used to measure the respirable concentration owing to the small quantities of dust sampled by the reference cyclone samplers. However, as expected all of the monitors were much more responsive to the respirable fraction of the airborne dust—the size of dust that they are primarily designed to measure. The Split 2, Dataram and Sidepak dust monitors were generally found to overestimate the respirable concentration and by varying degrees depending on the dust used, whereas the Sibata underestimated it.

The major potential advantage of the Split 2 dust monitor is the inclusion of the built-in IOM inlet and backup filter, which ideally should allow for calibration of the instrument without the need for a separate IOM reference sampler. However, in practice dust was found to leak around the cartridge containing the filter resulting in an underestimation of the true inhalable concentration. Attempts to eliminate leakage have been implemented but were only partially successful. The manufacturers have since redesigned the inlet with the aim of rectifying the problem.

The Respicon TM sampler does not display dust concentrations in real-time. Instead, the instrument gravimetrically measures the respirable, thoracic and inhalable dust concentrations, which are then used afterwards to calibrate the voltage output from three photometers that detect and measure the three size fractions. The Respicon was found to increasingly underestimate the gravimetric measurement of inhalable dust concentration (compared to the reference IOM sampler) as the particle size increased.

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